Sesha Charla
School of Mechanical Engineering,
Purdue University,
West Lafayette, Indiana, USA
email: scharla@purdue.edu

Peter H. Meckl[1]
Fellow ASME
School of Mechanical Engineering,
Purdue University,
West Lafayette, Indiana, USA
email: meckl@purdue.edu

Bin Yao
Fellow ASME
School of Mechanical Engineering,
Purdue University,
West Lafayette, Indiana, USA
email: byao@purdue.edu

Modelling and Parameter Estimation of Diesel Engine After-treatment System under Catalyst Saturation for Aging Diagnostics

The present work developes a dynamic model for NO_x reduction in diesel engine SCR systems when the catalyst is saturated with ammonia while explicitly considering the interplay between the residence time and the sampling frequency of the sensors. A novel parameter estimation algorithm is proposed utilizing the relationship between actual NO_x reduction and predicted NO_x reduction under saturation in the system while explicitly considering effects of the uncertainties due to NO_x sensor cross-sensitivity. Further, the model parameters are shown to be sensitive to the catalyst's aging state. Using the paramter estimates and their statistics, a hypothesis-testing framework based on the Wald test is developed to detect catalyst aging. The performance of the proposed aging detector is demonstrated using both test-cell data at different aging levels and real-world data from four long-haul trucks.

Keywords: Diesel Engine Aftertreatment System, System Identification, Nonlinear Systems, Hybrid Systems, Diagnostics, Parameter Estimation, Linear Programming.

1 Intorduction

Diesel engine after-treatment systems reduce the concentrations of harmful gases such as NO_x and CO from exhaust emissions. The Selective Catalytic Reduction (SCR) system reduces the engine-out NO_x into N_2 and H_2O, using ammonia from urea dosing in the presence of a catalyst. This catalytic conversion process is regulated to decrease the levels of ammonia in the exhaust (ammonia slip), through two methods. The first is feedback control, which adjusts the urea injection rate based on the exhaust NO_x concentrations. The second method involves an additional catalytic reaction, the Ammonia Slip Catalysis(ASC), which is designed to oxidize any excess ammonia at the end of the SCR bed. Figure-1 shows a schematic of the SCR-ASC system. It is found that NO_x-reduction capacity of the catalyst reduces with time due to hydro-thermal aging and deposition of contaminants such as excess urea, platinum, sulfur and phosphorous [1]. The aging of the catalyst leads to increased emissions of NO_x and ammonia slip, which can result in non-compliance with environmental regulations. A fault detection system for the aging of the catalyst would provide better control over the maintenance of the system and improve the overall reduction in emissions.

Thus, modern diesel after-treatment systems, particularly those that integrate Selective Catalytic Reduction (SCR) with Ammonia Slip Catalyst (ASC), necessitate advanced on-board diagnostics (OBD) tools for accurate assessment of catalyst aging. However, the effectiveness of traditional OBD approaches for this purpose has been impeded by the absence of model validation with real-world catalyst degradation data, the limitations imposed by existing commercial NO_x sensors' cross-sensitivity to ammonia and the absence of ammonia sensors in commercial vehicles.

Numerous studies have been conducted on modeling the SCR-ASC systems and their control [2]. A prevalent modeling approach is to approximate the PDE model for the plug flow reaction into a set of ODEs using sequence of Continuously Stirred Tank Reactors (CSTRs) approximation ([3] , and [4]). The single CSTR approach was first justified in [5] and a nonlinear model was developed using these assumptions, which was then linearized for feedback control design ([6]). With this model, observers were designed to estimate the states corresponding to the catalyst's storage ([7], [8]). A method for detecting the catalyst's aging using an observer for the the change in the maximum storage capacity of the catalyst, modeled as an exponential function of temperature, was also proposed in [7] and [9]. Alternatively, an onboad diagnostics (OBD) logic for catalyst aging based on the tailpipe NO_x and ammonia is proposed in [1] based on the hypthesis of reduction in adsorption cites and demonstrated on real world trucks. A common theme in these studies is that the system identification results in a non-convex, nonlinear parameter estimation problem. Moreover, these studies assume the availability of all the gaseous states at the tail-pipe for state estimation and to eliminate the effects of cross-sensitivity of the NO_x sensors, which is not always the case in real-world applications as tailpipe ammonia measurements are not

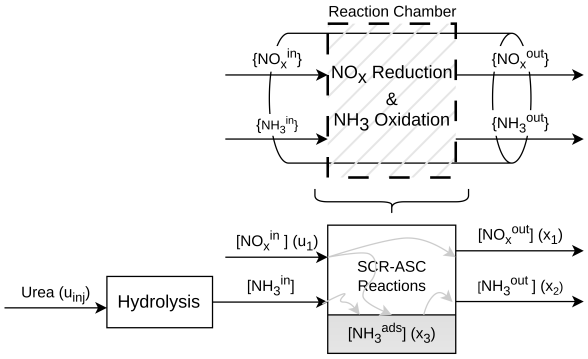


Fig. 1 Schematic of the SCR-ASC system

[1]Corresponding Author.
Version 0.1, December 12, 2025

generally available in commercial vehicles. One other fundamental issue with the CSTR model in the context of sensor sampling is that the discretization of the continuous model does not account for the resisdance time of the reactants resulting in high sampling frequency requirements that are not feasible in practice. Additionally, the discretization requires at least two CSTRs to capture the system dynamics and causality [10], thereby increasing the model order.

An alternative approach to the problem would be discarding the CSTR assumption and modelling the time evolution of the sensor signals when a "plug" or "parcel" of the exhaust gases flows through the chamber in discrete time considering the interplay between sampling and residence time of the reactants. The catalyst saturation that is inherently considered in the CSTR approach needs to be explicitly included as separate mode of the system. The present work developed such a model under catalyst saturation which becomes one of the modes of the complete switched nonlinear model. Following the model development we present the parameter estimation algorithm that estimates the parameters of the saturated catalyst model using the real-world data whose operating mode is unknown and has additional uncertainties due to NO_x sensor's cross-sensitivity to ammonia. Finally, we develope a statistical test for detecting the catalyst aging based on the estimated parameters of the saturated catalyst model. The performance of the proposed aging detector is demonstrated using both test-cell and real-world truck data.

2 SCR-ASC System Dynamics Under Catalyst Saturation

The SCR-ASC subsystem in the diesel engine after-treatment systems involves reacting flow on the top of a catalyst which adsorbs ammonia from urea decomposition. The reaction kinetics start with the decomposition of urea into ammonia followed by its adsorption onto the surface of the catalyst. The NO_x entering the chamber is reduced by this adsorbed ammonia. The Eley-Rideal reaction mechanism [11], [2], [3], [4] is considered for modeling the SCR reaction kinetics, where one reactant (NO_x) is gaseous, and the other is adsorbed on the catalyst surface (NH_3). [4] lists all the reactions that take place inside the SCR-ASC chamber. Ammonia oxidation (AMOX) happens both on the surfaces of the SCR catalyst and the ASC catalyst in the mechanism that is similar, but SCR catalyst favors NO_x reduction while ASC is specifically designed for ammonia oxidation. In order to keep the model order reasonably low and parameters identifiable, only the following three prominent reactions [5],[3] are considered:

1. Standard SCR reaction:

$$4NH_3^{ads} + 4NO + O_2 \xrightarrow{k_{scr}} 4N_2 + 6H_2O \qquad \{1\}$$

2. Ammonia Oxidation:

$$4NH_3^{ads} + 3O_2 \xrightarrow{k_{oxi}} 2N_2 + 6H_2O \qquad \{2\}$$

3. Ammonia Adsorption/Desorption:

$$NH_3 + \Theta_{free} \underset{k_{des}}{\overset{k_{ads}}{\rightleftharpoons}} NH_3^{ads} \qquad \{3\}$$

This aggregation of reactions results in errors in parameter estimates, specifically parameters containing rate constant for the NO_x reduction (k_{scr}) will have higher (bounded) uncertainty as it becomes dependent on the nitrogen selectivity of AMOX reaction [12]. Considering the reaction chamber as a control volume (Figure-2), the dynamic model is developed based on the molar conservation of species at the inlet and outlet separated by a residence-time interval. By assuming a zero-order hold on the inlet species over the sampling time, the equations relating inlet and outlet concentrations at the residence-time scale are extended to the sampling-time scale. The residence time can be parameterized as a function of mass flow rate (F) assuming that the effect of the density variation on residence time is insignificant.

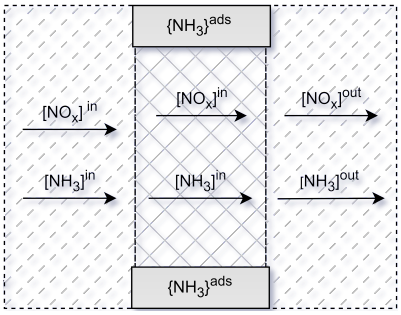


Fig. 2 Control volume in plug flow reaction

$$\tau(k) = \frac{\rho V_{scr}}{F(k)} = \frac{\tau_0}{F(k)} \qquad (1)$$

Let, $\{\bullet\}$ be the number of moles and $[\bullet]$ be the concentration of a given species $\bullet$. We have the molar conservation of NO_x across the control volume (V_{scr}) within one residence time (τ):

$$\{NO_x\}^{out}(t + (i+1)\tau) = \{NO_x\}^{in}(t + i\tau) + V_{scr}\int_0^{\tau} \frac{d}{dt}[NO_x]^{scr}dt \qquad (2)$$

The number of moles entering at the inlet and leaving at the outlet can be written as a function of the concentration, residence time and mass flow rate (F):

$$\{NO_x\}^{in/out} = [NO_x]^{in/out}\tau\left(\frac{F}{\rho}\right) = [NO_x]^{in/out}V_{scr} \qquad (3)$$

Using Eiley-Rideal mechanism for Standard SCR reaction {2}:

$$\frac{d}{dt}[NO_x]^{scr}dt = -k_{scr}[NH_3]^{ads}[NO_x]^{in} \qquad (4)$$

Rewriting (2) in terms of concentrations and integrating to residence time:

$$[NO_x]^{out}(t + (i+1)\tau) = [NO_x]^{in}(t + i\tau) - \tau k_{scr}[NH_3]^{ads}[NO_x]^{in} \qquad (5)$$

Summing the equations from (5) for $i = 0$ to $n-1$, where n is the total number of residence times within one sampling period $(n = t_s/\tau)$ and using the zero-order hold assumptions at the inlet (6) and outlet (7) concentrations, we get equation (8).

$$[NO_x]^{in}(t + i\tau) = [NO_x]^{in}(t) \quad \forall i \in [0, n-1] \qquad (6)$$

$$[NO_x]^{out}(t + (i+1)\tau) = [NO_x]^{out}(t + t_s) \quad \forall i \in [0, n-1] \qquad (7)$$

Let, tailpipe NO_x concentration be $x_1(k+1) = [NO_x]^{out}(t + t_s)$ and inlet NO_x concentration be $u_1(k) = [NO_x]^{in}(t)$, we get,

$$\underbrace{[NO_x]^{out}(t + t_s)}_{x_1(k+1)} = \underbrace{[NO_x]^{in}(t)}_{u_1(k)}$$

$$-\tau(t)k_{scr}(t)[NO_x]^{in}(t)\underbrace{\left\{\frac{1}{n}\sum_{i=0}^{n-1}[NH_3]^{ads}(t+i\tau)\right\}}_{\gamma(k)} \quad (8)$$

The quantity $\gamma(k)$ is the average concentration of ammonia adsorbed on the catalysts surface across the n residence times between k and $k+1$ samples. $\gamma(k)$ is bounded by the total concentration of voids on the surface of the catalyst, Γ, which decrease with increase in temperature.

$$0 \le \gamma(k) \le \Gamma \qquad \forall k \quad (9)$$

The unbounded dynamics of γ, $\gamma^{ub}(k)$ is a nonlinear function dependent on concentration of adsorbed ammonia, urea dosing (u_{inj}), and the inlet NO_x concentration from the previous time step.

$$\gamma^{ub}(k) = g_\gamma\left(\gamma(k-1), u_{inj}(k-1), [NO_x]^{in}(k-1)\right) \quad (10)$$

The actual expression, which is beyond the scope of this work, can be derived using the similar approach used for deriving NO_x process dynamics (8). $\gamma(k)$ can be described using the following piecewise function:

$$\gamma(k) = \begin{cases} \gamma^{ub}(k) & \text{if} \quad 0 \le \gamma^{ub}(k) \le \Gamma \\ \Gamma & \text{if} \quad \gamma^{ub}(k) > \Gamma \quad \text{[Satuaration]} \\ 0 & \text{if} \quad \gamma^{ub}(k) < 0 \quad \text{[Empty]} \end{cases} \quad (11)$$

Let, $\eta(k)$ be the NO_x reduction during a given time step k,

$$\eta(k+1) = u_1(k) - x_1(k+1) \quad (12)$$

Introducing above definition and the residence time expression (1) into the NO_x process dynamics model (8), we get,

$$\eta(k+1) = \frac{u_1(k)}{F(k)}\tau_0\ k_{scr}(T(k))\ \gamma(T(k)) \quad (13)$$

For NO_x process dynamics under catalyst saturation, $\gamma(k) = \Gamma$.

$$\implies \eta_{sat}(k+1) = \frac{u_1(k)}{F(k)}\tau_0\ k_{scr}(T(k))\ \Gamma(T(k)) \quad (14)$$

The above equation (14) shows that the tailpipe NO_x concentration becomes independent of urea dosing under catalyst saturation. Polynomial approximations of Arrhenius temperature dependence of the rate constant, k_{scr} and the concentration of viable voids, Γ ([4],[13],[14]), are used to get a linear in parameters model for the NO_x process dynamics.

$$k_{scr}(T) = A_{scr}e^{\left(-\frac{E_{scr}}{RT}\right)} \approx \sum k_i T^i \qquad i = 0, 1, \ldots \quad (15)$$

$$\Gamma(T) = S_1 e^{-S_2 T} \approx \sum r_i T^i \qquad i = 0, 1, \ldots \quad (16)$$

The above approximate models for temperature introduces dependence of the parameter estimates based on temperature range in the samples. For numerical stability [15], the given temperature range in mapped to $[-1, 1]$ and chebyshev polynomial basis [16] are used instead of standard polynomials. Incorporating equations (1) and (16) into the NO_x process dynamic model (14),

$$\eta_{sat}(k+1) = \phi_{sat}^T(T(k))\ \theta_{sat} \quad (17)$$

$$\phi_{sat}(k) = \frac{u_1(k)}{F(k)}\left[2\left(\frac{T-T_0}{T_r}\right)^2 - 1 \quad \frac{T-T_0}{T_r} \quad 1\right] \quad (18)$$

Where, $T_0 = \frac{T_{max}+T_{min}}{2}$, $T_r = \frac{T_{max}-T_{min}}{2}$

It can be observed that, in a given time step maximum NO_x reduction happens when the catalyst is saturated. Thus, under the same temperature, flow rate and inlet NO_x concentration, the response of the saturated model (14) forms a tight upper bound to the response of the actual system, that switches between saturation and desaturation based on urea dosing, flow rate and catalyst temperature. This results in the following bounding condition (19) on response of the saturated system.

$$\eta_{sat}(k) \ge \eta(k) \qquad \forall k \quad (19)$$

This relationship is used for model parameter estimation.

2.1 NO_x sensor cross-sensitivity

Commercial NO_x sensors are cross-sensitive to tailpipe ammonia.The resulting error is modeled as a function of ammonia concentration and a temperature-dependent cross-sensitivity factor $\chi(T)$. This introduces a non-negative directional error $(\varepsilon_\chi = \chi(T(k)) \times [NH_3]^{out} \ge 0)$ into the tailpipe NO_x measurement y_1. Accordingly, when the NO_x reduction in a time-step is calculated from sensor measurements, $y_1(k) = x_1(k) + \varepsilon_\chi(k)$, the estimated quantity less than the actual NO_x reduction.

$$\eta_y(k) = u_1(k-1) - y_1(k) = \eta(k) - \varepsilon_\chi(k) \le \eta(k) \quad (20)$$

Thus, the bounding condition on the saturated system's response (19) remains valid when the NO_x reduction during the time-step is estimated from the NO_x sensor measurements.

$$\eta_{sat}(k) \ge \eta(k) = \eta_y(k) + \varepsilon_\chi(k) \ge \eta_y(k) \quad (21)$$

The FTIR sensor also has a bias that is directional (≥ 0). Simlar

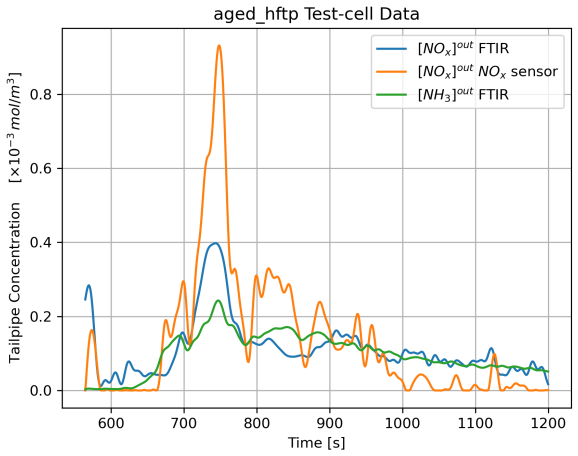


Fig. 3 NO_x sensor cross-sensitivity to tailpipe ammonia

arguements as above can be made for the validity of the bounding condidtion.

3 Maximum Likelihood Estimation of the Model Parameters

The catalyst switches between saturation mode and normal mode based on states that are not completely controllable experimentally. The bounding condition on the saturated system's response (19) can be used for detecting segments of the data where the catalyst's operation is normal or saturated. As the saturated catalyst's response forms a tight upper bound to the actual system's response,

the parameters of the systems will be the solution to the linear programming problem that minimizes the area under the curve with the lower bound as the response of the actual system.

$$\text{min.} \quad \sum \eta_{sat}(k) \quad \text{s.t} \quad (\eta_{sat}(k) \geq \eta(k)) \quad \forall k \tag{22}$$

The optimization problem (22) is feasible as $\eta_{sat}, \eta \geq 0$ always. When the NO_x sensor data is used, $\eta(k)$ is replaced with $\eta_y(k)$ as the bounding condition still holds (21) . Rewriting the optimization problem using regression vector (17), we have the linear programming problem for estimating the parameters of the catalyst:

$$\begin{aligned} \text{minimize} \quad & \mathbf{1}^T \left[\Phi_{sat}\theta_{sat}\right] \\ \text{subject to} \quad & \Phi_{sat}\theta_{sat} \geq H, \\ \text{where,} \quad & \Phi_{sat} = \left[\phi_{sat}(1), \phi_{sat}(2), \ldots, \phi_{sat}(N-1)\right]^T \\ & H = \left[\eta(2), \eta(3), \ldots, \eta(N)\right]^T \end{aligned} \tag{23}$$

$\mathbf{1}$ is a column vector of $1's$.

The above estimate of θ_{sat} is the ***maximum likelihood estimate (MLE) under half-normal distribution [17] for model structure error***, (ε_η), between the saturated system's response, $\eta_{sat}(k)$, and the actual response of the catalyst, $\eta(k)$ (Appendix-A). In the case of NO_x-sensor data with cross-sensitivity, the model structure error includes the cross-sensitivity error ε_χ and is assumed to be an half-normal random variable.

Using asymptotic properties of MLE (Theorem-7.3 in [18]), with significantly large data length (N), the parameter estimate $\hat{\theta}_{sat}$ asymptotically follows a normal distribution under some regularity conditions on the PDF $p(\boldsymbol{\varepsilon}_n; \theta_{sat})$.

$$\hat{\theta}_{sat} \sim \mathcal{N}\left(\theta_{sat}, I^{-1}(\theta_{sat})\right) \tag{24}$$

where, $I(\theta_{sat})$ is the fisher information matrix evaluated at the true value of the parameter (θ_{sat}). The regularity conditions require the existence of derivatives of the log-likelihood function and the Fisher information being non-zero. Using the log-likelihood function for half-normal distribution (A4), we have,

$$[I(\theta_{sat})]_{ij} = -E\left[\frac{\partial^2 L(\theta_{sat})}{\partial\theta_j \partial\theta_i}\right] = \frac{1}{\sigma^2}\sum_{k=1}^{N}\frac{\partial\varepsilon_\eta(k)}{\partial\theta_i}\frac{\partial\varepsilon_\eta(k)}{\partial\theta_j}$$

$$I(\theta_{sat}) = \frac{1}{\sigma^2}\left[\Phi_{sat}^T\Phi_{sat}\right] \tag{25}$$

The variance in the η_{sat} prediction for the given time-step is obtained by propagating the uncertainity of paramter estimates (24). Using, (17), we have,

$$\text{Var}\left[\eta_{sat}(k)\right] = \phi_{sat}(k)^T I^{-1}(\theta_{sat})\phi_{sat}(k) \tag{26}$$

The parameter estimates using FTIR measurements from the test-cell experiments are tabulated in Table-1. The parameters show consistency across repeated RMC (Ramped Mode Cycle) and hot-FTP (Federal Test Procedure) tests on the same catalyst condition validating the estimation approach. However, the parameters vary significantly between RMC and hot-FTP tests. This is predominatly due to the differences in the temperature ranges experienced by the catalyst during the tests, which affects the Arrhenius temperature dependence of the rate constant and the concentration of viable voids on the catalyst surface. The temperature range of the data considered during the RMC tests is between 240°C and 360°C, while for the Hot-FTP tests it is between 200°C and 300°C.

Table 1 Parameter Estimates using FTIR Data

Test	$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\theta}_3$	σ
DG RMC 1	-0.63	0.55	38.74	1.14
DG RMC 2	-0.82	0.69	38.61	1.08
DG RMC 2	-0.80	1.12	38.78	1.10
Aged RMC	-1.31	1.89	36.94	1.01
DG Hot FTP 1	-2.14	-5.80	39.53	1.35
DG Hot FTP 2	-3.45	-7.99	38.05	1.30
DG Hot FTP 3	-3.37	-6.42	38.88	1.33
Aged Hot FTP	-4.54	-8.01	37.00	1.31

The parameter estimates using NO_x sensor measurements from the test-cell experiments are tabulated in Table-2. The parameter estimates are different from those obtained using FTIR measurements (Table-1) due to the cross-sensitivity of the NO_x sensor to ammonia. However, the parameters show consistency across repeated tests on the same catalyst under similar test conditions. The NO_x sensor cros -sensitivity error $\varepsilon_\chi(T)$ introduces bias in the paramter estimates as part of the error that has the same temperature dependence as the model structure enters into the parameter estimates. Thus, the linear programming problem estimated the paramters with an added bias term as shown below:

$$\varepsilon_\chi(T) = \phi_{sat}(T(k))^T\theta_{\varepsilon_\chi} + \epsilon_\chi \tag{27}$$

$$\begin{aligned} J(\theta_{sat}) &= \sum_{k=0}^{N-1}\left(\phi^T(k)\theta_{sat} - \eta_y(k)\right) \\ &= \sum_{k=0}^{N-1}\left(\phi^T(k)\left(\theta_{sat} + \theta_{\varepsilon_\chi}\right) - \eta(k)\right) \end{aligned} \tag{28}$$

Thus, the estimated parameters using NO_x sensor measurements will have a bias dependent on the cross-sensitivity error characteristics of the sensor used. From, the experimental results, it is observed that NO_x sensor cross-sensitivity reduces the difference between the estimates of aging-factor (Figure 6, 7) between aged and degreened catalysts. But the overall trend of change in parameter values remains consistent.

Table 2 Parameter Estimates using NO_x Sensor Data

Test	$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\theta}_3$	σ
DG RMC 1	-0.70	0.56	38.48	1.10
DG RMC 2	-1.18	0.93	38.43	1.07
DG RMC 3	-0.54	0.62	38.99	1.06
Aged RMC	-1.03	1.26	37.80	1.00
DG Hot FTP 1	-2.30	-6.11	37.08	1.28
DG Hot FTP 2	-2.71	-6.13	36.22	1.27
DG Hot FTP 3	-3.92	-6.17	35.81	1.30
Aged Hot FTP	-6.42	-13.06	34.73	1.34

Furthermore, the above delineated parameter estimation method has a limitation that the data-set must have response of the catalyst under saturation for finite duration. If this condition is not met, the algorithm estimates the paramters considering the maximum concentration of adsorbed ammonia as the concentration corresponding to catalyst saturation (Appendix-B).

4 Catalyst Mode Detection

The saturated catalyst model's response is not dependent on the previous state but only on the inputs making the response independent of the initial conditions. Thus, using the parameter estimates, a given data point can be classified as saturated catalyst's response

if the prediction error for the same inputs (inlet NO_x, u_1, and mass flow rate, F) is less than the ϵ_{sat}.

$$|\hat{\eta}_{sat}(k) - \eta(k)| \leq \epsilon_{sat} \implies \text{Catalyst Saturation} \quad (29)$$

For the current test-cell and truck data set, $\epsilon_{sat} = 2.5 \times 10^{-3}\, mol/m^{-3}$.

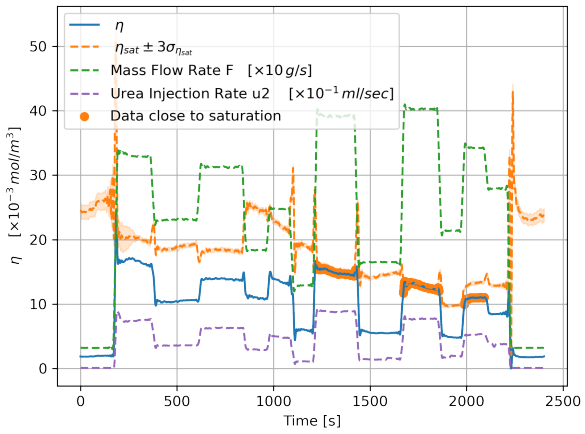


Fig. 4 Catalyst Mode Detection for DG RMC 1 Data

The catalyst mode detection for DG RMC 1 test-cell data using FTIR measurements is shown in Figure-4. The catalyst is found to be close to saturation (or at its maximum NO_x reduction) when the mass flow rate and the urea dosing rates are high, consistent with the physical understanding of the catalyst operation. Further, the flow rate can introduce ~~virtual cieling~~ on the maximum achievable NO_x reduction due to reduced residence time in the catalyst which would be indistinguishable from saturated catalyst behaviour as the NO_x reduction would be independent of urea dosing.

A fraction of the data from the test-cell experiments under RMC and Hot-FTP test conditions for both degreened and aged catalysts are found to be in saturated mode. The number data points close to catalyst saturation (N_{sat}) and the total number of samples (N) for different test conditions is tabulated in Table-3, 4. A higher

Table 3 Data Points Close to Catalyst Saturation in Test-Cell FTIR Data

Test	N	N_{sat}
DG RMC 1	2402	490
DG RMC 2	2402	493
DG RMC 3	2402	491
Aged RMC	2402	515
DG Hot FTP 1	634	186
DG Hot FTP 2	634	181
DG Hot FTP 3	638	180
~~Aged Hot FTP~~	~~636~~	~~198~~

number of data points classified as close to catalyst saturation in RMC test conditions compared to Hot-FTP, this results in better parameter estimates and performace of the aging detection test statistic (Table-5).

5 Model Paramters Capture Catalyst Aging

The paramters of the saturated system are a function of the rate-constant for SCR reaction and the maximum surface concentration of vaible voids on the catalyst. The adsorption cite concentration is

Table 4 Data Points Close to Catalyst Saturation in Test-Cell NO_x Sensor Data

Test	N	N_{sat}
DG RMC 1	2402	492
DG RMC 2	2402	494
DG RMC 3	2402	493
Aged RMC	2402	515
DG Hot FTP 1	634	194
DG Hot FTP 2	634	191
DG Hot FTP 3	638	198
Aged Hot FTP	636	220

hypothesized to change with aging of the catalyst allowing the parameters to capture the aging of the catalyst. This is demonstrated using the test-cell experiments on a new (de-greened) and hydrothermally aged catalyst with the same capacity. The predicted NO_x reduction for an aged catalyst is found to be statistically lower than the predicted NO_x reduction in a de-greened catalyst (Figure-5). When the η_{sat} response is nromalized with respect to the inlet

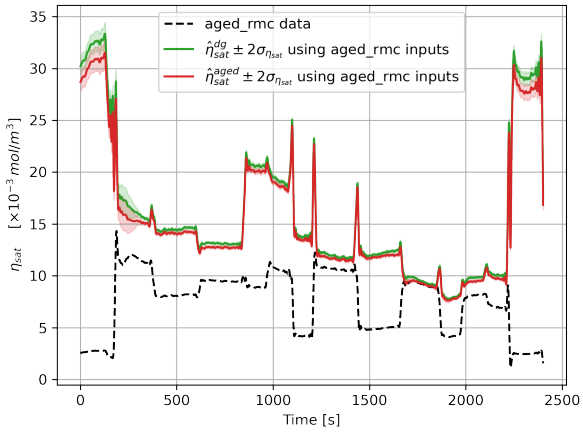


Fig. 5 Predicted η_{sat} for Aged and Degreened Catalyst

NO_x concentration and flow-rate, we end up with a quantity that is only a function of themperature and represents the product of rate-constant and maximum surface coverage. From 17,

$$\alpha_{sat} = \frac{F(k)}{u_1(k)}\eta_{sat}(k) = \tau_0 k_{scr}\Gamma = \phi_T^T(k)\theta_{sat} \quad (30)$$

$$\text{where,} \quad \phi_T^T(k) = \begin{bmatrix} 2\left(\frac{T-T_0}{T_r}\right)^2 - 1 & \frac{T-T_0}{T_r} & 1 \end{bmatrix} \quad (31)$$

This quantity, α_{sat} is estimatible and changes with aging of the catalyst based on the hypothesis of changing adsorption cite concentration. The change in α_{sat} with aging is demonstrated using the test-cell experiments (Figure 6, 7).

Thus, θ_{sat} capture the aging of the catalyst. Hence, the catalyst aging detection problem can be posed as a hypothesis testing problem with null hypothesis ($\mathcal{H}_0$) corresponding to a degreened catalyst and alternative hypothesis ($\mathcal{H}_1$) corresponding to an aged catalyst. Since, the distribution of the error is assumed to be known (half-normal) for both hypotheses and differ by the unknown (but estimatible) parameter θ_{sat}, the following Wald test [19] is proposed to check if the parameter estimates for the given data set correspond to an aged catalyst or a known degreened catalyst. Decide $\mathcal{H}_1$ if,

$$T_w = \left(\hat{\theta}_{sat} - \theta^{dg}_{sat}\right)^T I\left(\hat{\theta}_{sat}\right)\left(\hat{\theta}_{sat} - \theta^{dg}_{sat}\right) > \beta \quad (32)$$

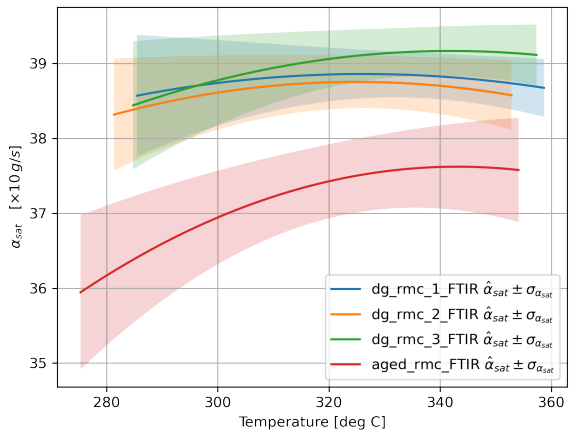


Fig. 6 $\hat{\alpha}_{sat}$ from RMC FTIR Data

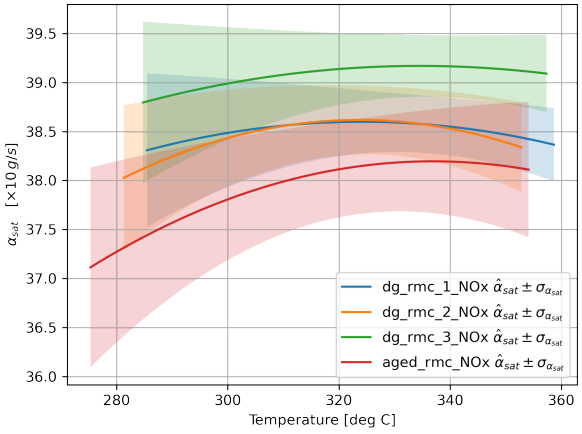


Fig. 7 $\hat{\alpha}_{sat}$ from RMC NO_x Sensor Data

Where, θ_{sat}^{dg} is the parameter estimates for the degreened catalyst under same input conditions i.e., temperature, flow-rate and urea-dosing rate ranges. The threshold β is decided based on the experimental results. For RMC and FTP data sets, the value of the test-statistic is tabulated in Tabel 5. The distribution of the test-statistic under aging (non-central χ^2, asymptotically) is different for RMC and FTP test conditions due to the differences in the ranges of inputs (temperature, flow-rate and urea-dosing rate) experienced by the catalyst during the tests. Based on the results, a reasonabley discerning threshold for Hot-FTP and RMC test conditions is found to be:

$$\beta_{hFTP} = 4.00, \qquad \beta_{RMC} = 10.00 \tag{33}$$

Further investigation with more data is needed to arrive at the probability of detection and false alarm for these thresholds. The proposed Wald test detects the aged catalyst for both FTIR and NO_x sensor measurements under both test conditions.

6 Aging Detection from Truck Data

The performance of the proposed aging detector is also demonstrated using the NO_x sensor data from four long-haul truck. The road data was collected from trucks operating in the United States for two days separated by a few years (Table 6) during which the catalyst has aged based on the existing performance measures. The

Table 5 Wald Test Results On the Test Cell Data

Test	T_w from FTIR	T_w from NO_x sensor
DG RMC 1	0.26	5.54
DG RMC 2	0.00	0.00
DG RMC 2	6.71	5.46
Aged RMC	47.16	17.68
DG Hot FTP 1	0.49	0.81
DG Hot FTP 2	3.26	1.98
DG Hot FTP 3	0.00	0.00
Aged Hot FTP	4.84	11.44

data contains measurements of NO_x sensors before and after the catalyst, as well as other required variables including flow rate, temperature and urea injection rate. The data preporcessing includes, 1. removing the rows with missing values, 2. removing the rows whose operating temperature range is beyond range of $200 - 300\,^0C$, 3. interpolating the missing data if the data breaks are smaller than a minute, and finally, 4. smooting the data using non-causal chebyshev filter. The preprocessed data is than partitioned into drive segments with no breaks which correspond to a continuous driving period from engine start to engine stop. Un-

Table 6 Truck Data Set Years and Mileage Difference

Truck	Earlier Data Year	Latter Data Year	Milage Difference
AD Transport (adt)	2015	2017	2.79×10^5
Mesilla Valley (mes)	2015	2018	11.87×10^5
Werner (wer)	2015	2017	5.89×10^5
Transwest (trw)	2015	2016	6.56×10^5

like the test-cell data where the engine drive-cycles are repeatable and produce significant fraction of samples where the catalyst is close to saturation, the truck data drive segments constitute varied driving conditions with fewer samples $\{\leq 10\%\}$ that are close to catalyst saturation. A necessary conditon for the proposed parameter estimation and aging detection framework is that the data contains sufficiently large number of samples close to catalyst saturation such that the asymptotic properties of the MLE and the test statistic holds (34). Hence, the drive segment data is further pruned to only include segments with atleast 150 samples (120 in the case of Werner) close to catalyst saturation.

$$N_{sat} \geq N_{sat}^{min} \tag{34}$$

$$\beta_{truck} = 250 \tag{35}$$

The threshold of the Wald test statistic (35) for aging detection in truck is chosen such that it maximises the probability of detection while minimizing the probability of false alarm and missed detection for the given limited truck data. A larger data set would help arriving at the thresholds using Neyman-Pearson criteria [20]. The value of the test statistic for the drive segments for each truck is tabulated in Tables 8, 10, 11 and 9. A summary of the aging detection results is tabulated in the Table 7. The detector performs

Table 7 Summary of Aging Detection Results on Truck Data

Truck	Total Drive Segments	Detection	False Alarm	Missed Detection
AD Transport	8	4	0	4
Mesilla Valley	11	9	1	1
Werner	9	7	1	1
Transwest	8	6	2	0

reasonably well on three of the four trucks with high probability

of detection and low false alarm and missed detection rates. The AD Transport truck has relatively lower milage difference between the earlier and latter data years which may explain the lower detection performance. The performance is limited by the amount of data available with sufficient samples close to catalyst saturation. Further, the lack of ground truth on the actual catalyst aging levels limits the ability to quantify the performance of the detector. Nevertheless, the results demonstrate the potential of the proposed aging detection framework for on-board application using NO_x sensor data.

7 Conclusion

The dynamics of NO_x reduction in diesel engine SCR systems under catalyst saturation were used to develop an aging detector based on the hypothesis that the model parameters associated with catalyst storage capacity change as the catalyst ages. A dynamic model for NO_x reduction under saturation was formulated from molar conservation while explicitly accounting for the interplay between the residence time of the reacting flow and the sampling time of the sensors. A maximum-likelihood parameter estimation algorithm was proposed using the bounding condition between the actual system response and the response under catalyst saturation. The algorithm was validated using experimental data from test-cell under RMC and hot-FTP conditions. The NO_x-sensor's ammonia cross-sensitivity was shown to introduce a bias in the parameter estimates, reducing the separation between the estimates for degreened and aged catalysts. A hypothesis-testing framework based on the Wald test was then developed for catalyst aging detection using the estimated parameters. The detector's performance was demonstrated using test-cell data at different aging levels and real-world data from four long-haul trucks. The results indicate that the proposed framework can reliably identify catalyst aging even in the presence of sensor cross-sensitivity and varying operating conditions, provided that the data include a sufficient number of samples near the saturation regime, which forms a necessary condition for aging detection.

Acknowledgment

Authors acknowledge the support and feedback from David Schmidt, Larry Bruner, Adam Kidd and Tom Nelson of Cummins, Inc.

Funding Data

Authors acknowledge the funding and test-cell, truck and simulation data of Diesel Engine ~~After-treatment~~ Systems from Cummins, Inc.

Nomenclature

Symbols

$\{\bullet\}$ = Number of moles of $\bullet$. $(moles)$
$[\bullet]$ = Concentration of $\bullet$. (mol/m^3)
u_1 = Engine-out (SCR-in) NO_x concentration
x_1 = Tailpipe (SCR-out) NO_x concentration
x_1 = Tailpipe (SCR-out) NO_x sensor measurement
Γ = Surface concentration of the total storage capacity for the catalyst (mol/m^2)
E_{scr} = Activation Energy of SCR reaction
A_{scr} = Pre-exponential factor of SCR reaction
R = Universal gas constant
T = Temperature
k_i = Coefficients of polynomial temperature model for k_{scr}
S_1, S_2 = Coefficients of the temperature model for Γ, $S_1, S_2 \geq 0$
r_i = Coefficients of polynomial temperature model for Γ
V_{scr} = Volume of the catalyst chamber
F = Mass flow rate of the exhaust gases (g/s)
ρ = Density of the exhaust gases
$\gamma(k)$ = Average surface concentration of adsorbed ammonia in the sample [k] to [k+1]
ε_χ = Error in NO_x due to ammonia cross-sensitivity
χ = Cross-sensitivity factor
ε_η = Model structure error between saturated catalyst model and actual system
$\phi_{sat}(T)$ = Regression vector for saturated catalyst model
θ_{sat} = Parameter vector for saturated catalyst model
η_{sat} = NO_x reduction per time-step under catalyst saturation
σ_η = Standard deviation in NO_x
σ = Scale parameter for half-normal distribution
$I(\theta)$ = Fisher Information Matrix
N_{sat} = Number of data points close to catalyst saturation
N = Total number of data points
N_{sat}^{min} = Minimum number of data points close to catalyst saturation for reliable parameter estimation
T_w = Wald test statistic for catalyst aging detection
β = Threshold for Wald test statistic for catalyst aging detection
θ_{sat}^{dg} = Parameter estimates for degreened catalyst
α_{sat} = Flow and inlet NO_x normalized NO_x reduction under catalyst saturation

Abbreviations

SCR = Selcetive Catalytic Reduction
ASC = Ammonia Slip Catalysis
CSTR = Continuously Stirred Tank Reactor
PFR = Plug Flow Reactor
RMC = Ramped Mode Cycle
FTP = Federal Test Procedure
MLE = Maximum Likelihood Estimate
PDF = Probability Density Function
adt = AD Transport
mes = Mesilla Valley
wer = Werner
trw = Transwest

Appendix A: Maximum Likelihood Estimate Derivation

Consider,

$$\varepsilon_\eta(k) = \eta_{sat}(k) - \eta(k) \quad \Longrightarrow \quad \varepsilon_\eta(k) = \left(\phi_{sat}^T \theta_{sat} - \eta(k)\right) \tag{A1}$$

based on the measurement sensors. Assuming ε_η follows half-normal distribution [21] with scale-parameter, σ,

$$\varepsilon_\eta \backsim \text{Half-Normal}(\sigma); \quad p(\varepsilon_\eta;\sigma) = \begin{cases} \frac{\sqrt{2}}{\sigma\sqrt{\pi}} e^{-\frac{\varepsilon_\eta^2}{2\sigma^2}} & \varepsilon_\eta \geq 0 \\ 0 & \varepsilon_\eta < 0 \end{cases} \tag{A2}$$

we have the likelihood and log-likelihood functions given the data and parameter estimate θ_{sat} and σ:

$$l(\theta_{sat}) = p(\boldsymbol{\varepsilon}_\eta;\theta_{sat},\sigma) = \prod_{k=1}^{N} \frac{\sqrt{2}}{\sigma\sqrt{\pi}} e^{-\frac{\varepsilon_\eta^2}{2\sigma^2}} \quad \varepsilon_\eta(k) \geq 0 \tag{A3}$$

$$L(\theta_{sat}) = \ln(l(\theta_{sat})) = \frac{N}{2}\ln\frac{2}{\sigma^2\pi} - \frac{1}{2\sigma^2}\sum_{k=1}^{N}\varepsilon_\eta^2 \quad \varepsilon_\eta(k) \geq 0 \tag{A4}$$

where, N is the total sample size. The maximum likelihood estimate for θ_{sat} would be:

$$\hat{\theta}_{sat} = \arg\max_{\theta_{sat}} \left\{ \frac{N}{2}\ln\frac{2}{\sigma^2\pi} - \frac{1}{2\sigma^2}\sum_{k=1}^{N}\varepsilon_\eta^2 \right\} \quad \varepsilon_\eta(k) \geq 0$$

$$= \arg\min_{\theta_{sat}} \left(\sum_{k=1}^{N} \varepsilon_{\eta}^{2} \right) \quad \varepsilon_{\eta}(k) \geq 0$$

$$= \arg\min_{\theta_{sat}} \left(\sum_{k=1}^{N} \left(\phi_{sat}(k)^{T} \theta_{sat} - \eta(k) \right)^{2} \right)$$

$$s.t \quad \phi_{sat}(k)^{T} \theta_{sat} \geq \eta(k) \quad \forall\, k \in 1, 2, \ldots, N \qquad \text{(A5)}$$

As the inequality constraint ensures that the cost function and individual elements of the sum are always positive. Thus, the l_2-norm in quadratic programming problem (A5) can be replaced with l_1-norm resulting an linear programming problem formulation (A6).

$$\hat{\theta}_{sat} = \arg\min_{\theta_{sat}} \left(\sum_{k=1}^{N} \left(\phi_{sat}(k)^{T} \theta_{sat} \right) \right)$$

$$s.t \quad \phi_{sat}(k)^{T} \theta_{sat} \geq \eta(k) \quad \forall\, k \in 1, 2, \ldots, N \qquad \text{(A6)}$$

Appendix B: Limitation: Parameter Estimation Assumes Catalyst Saturation

The catalyst mode detection algorithm presented fails when the catalyst is not saturated within the duration for a significant length of time in the data. If the data-set does not have any regions of saturation, the approach fits a saturated model as if the maximum achieved catalyst ammonia storage, $(\max(\sigma(k)))$, is the storage capacity of the catalyst (Γ). This is demonstrated using data from high-fidelity simulation model (AVL Cruise [22]) tuned to RMC (Ramped Mode Cycle) test on a test-cell. The RMC test is then simulated with ±20% change in the gain of urea-dosing controller to change the length of the data where the catalyst remains saturated. Figure-8 shows no increase in NO_x reduction when urea

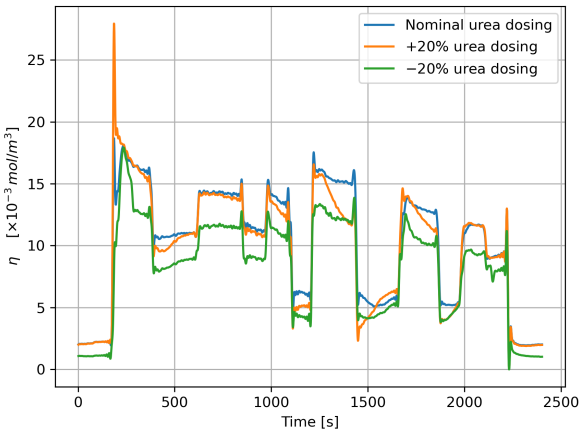


Fig. 8 RMC η simulation with ±20% Urea Dosing

dosing is increased by 20%, which indicates that maximum NO_x reduction is already reached at nominal urea dosing. This could be because the catalyst is saturated or because the inlet NO_x concentration is sufficiently low. Conversely, a 20% decrease in urea dosing decreases NO_x reduction. When the −20% urea-dosing data are used to estimate saturated-model parameters, the model predicts a substantially lower NO_x reduction under saturation than for the nominal and +20% dosing cases (Figure-9), the latter two giving essentially identical results. Thus for the nominal and +20% dosing cases, the catalyst is indeed saturated, and the parameter estimates show the same. However, for the −20% dosing case, the

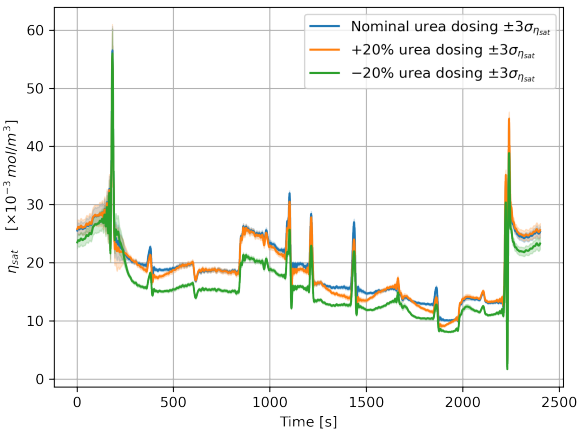


Fig. 9 RMC η_{sat} prediction with ±20% Urea Dosing

catalyst is not saturated in the data, instead the parmeters fit the maximimu surface coverage achieved $\{\theta_{max} = {}^{\gamma_{max}}/_{\Gamma}\}$ in the data, i.e.,

$$\eta_{sat}(k+1) = \frac{u_1(k)}{F(k)} \tau_0\, k_{scr} \Gamma \theta_{max} = \phi_{sat} \theta_{sat} \qquad \text{(B1)}$$

It is assumed that under ordinary drive conditions, the catalyst will reach saturation for a significant length of time, making the proposed parameter estimation approach applicable.

Appendix C: Aging Detector Results on Truck Data Drive Segments

The value of the Wald test statistic for each drive segment that met the necessary condition of sufficient samples close to catalyst saturation is calculated and tabulated in Tables 8, 10, 11 and 9 for each truck. The number and percentage of samples close to catalyst saturation in each drive segment is also tabulated. There is a strong corrolation between detection of aging and the number of samples close to catalyst saturation in the drive segment.

Table 8 Aging Detaction Results for AD Transport

Drive Segment	T_w	N_{sat}	% $({}^{N_{sat}}/_{N})$
adt_15_0	0.00	797	6.70
adt_15_1	28.95	305	2.75
adt_15_2	58.50	276	5.93
adt_15_3	89.56	271	12.77
adt_17_0	65.59	414	2.97
adt_17_1	127.79	614	6.76
adt_17_2	12.13	330	8.44
adt_17_3	61.76	336	9.73

Table 9 Aging Detaction Results for Transwest

Drive Segment	T_w	N_{sat}	% $({}^{N_{sat}}/_{N})$
trw_15_0	0.00	214	1.29
trw_15_2	98.99	160	2.95
trw_15_4	1546.55	213	4.90
trw_15_6	935.05	152	8.86
trw_16_1	423.20	246	3.67
trw_16_2	1022.06	199	5.22
trw_16_3	697.78	170	5.36
trw_16_5	313.62	200	9.31

Table 10 Aging Detaction Results for Mesilla Valley

Drive Segment	T_w	N_{sat}	% ($^{N_{sat}}/_{N}$)
*mes*_15_0	672.43	167	1.26
*mes*_15_1	106.66	189	1.81
*mes*_15_3	237.79	159	2.95
*mes*_15_4	0.00	226	6.03
*mes*_15_6	84.73	179	5.75
*mes*_18_0	543.55	231	2.10
*mes*_18_1	1050.24	308	3.00
*mes*_18_4	22.48	195	6.14
*mes*_18_6	7150.19	378	12.73
*mes*_18_7	760.89	224	8.14
*mes*_18_8	342.67	246	9.04

Table 11 Aging Detaction Results for Werner

Drive Segment	T_w	N_{sat}	% ($^{N_{sat}}/_{N}$)
*wer*_15_1	0.000	186	1.30
*wer*_15_4	113.86	143	3.84
*wer*_15_5	4340.08	124	4.16
*wer*_17_0	700.88	186	1.24
*wer*_17_2	3993.85	446	7.23
*wer*_17_3	1693.49	199	4.00
*wer*_17_4	186.64	120	2.81
*wer*_17_7	1878.52	129	4.73
*wer*_17_9	7500.10	218	11.63

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